Investigation report

Can we trust acquisition-channel conclusions when source values are missing?

Assess the effect of missing acquisition source on channel conclusions.

Data

Authorized PocketToons SQLite
April-September 2026, UTC

Snapshot

Users snapshot 2026-09-30 23:59:59 UTC
Build 3a0dbed21d2f45958aefa2887a0cadd8

Run

Independent
2026-11-09
14 logged checks

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How to read this report

- Check numbers refer to individual JSONL query logs in this folder.
- Unknown is an unallocated first-touch source; hypothetical reassignment is a sensitivity bound, not a correction.

Terms used in this report

Term	Meaning
Signup	One registered user counted in the UTC month of signup.
New payer	A user counted in the month of their first successful charge, by their recorded acquisition source.
MAU	Distinct identified users active in a UTC month.

01. Chapter

The answer

01

Qualified yes for large descriptive differences in this fixture: missing sources are approximately 1% of signups, and the example channel shift remains large under extreme reassignment. This does not establish that recorded sources are correct or that marketing caused the outcomes.

- Across April-September, 135 of 13,600 signups (0.99%) have unknown source. Monthly unknown share ranges from 0.98% to 1.01% (checks 5 and 10).
- In July, paid social has 1,582 signups versus 453 organic; allocating all 25 unknown signups to organic still leaves social ahead by 1,104 (check 14).
- The database build registry hash differs from the current repository registry hash; verify the precise definition changes before relying on Gold for consequential decisions (check 4 and run notes).

02. Chapter

How we investigated

02

What data we used

Item	Definition and coverage
Period	Complete UTC months [2026-04-01, 2026-10-01); monthly signup and first-payment event counts.
Definitions	Supporting metrics v1.1; MAU v1.1. Source is the fixed first-touch user label. Unknown remains separate. Same-month CAC is spend divided by first-time payers in the month, not cohort CAC.
Lineage	Gold marketing and activity marts for compatible monthly aggregates, Silver profile for checks, Bronze users for raw values.
Build	Succeeded 2026-11-09; build-time reconciliation passed. Recorded source fingerprints have not been recomputed. Snapshot is metadata, not a verified hash.

Investigation path

- Baseline: inventory layers, build, population, unknown labels and monthly segment counts (checks 1-13).
- Round 1: test whether unknown attribution is material; whether it can overturn illustrative channel comparisons; and whether recorded source proves causal attribution (checks 5, 10, 14).
- Stopped after one round: illustrative sensitivity is bounded; label accuracy and causality cannot be established from these fields alone.

03. Chapter

What happened

03

The movement

Month	Unknown signups / total; unknown active / MAU
Apr	16 / 1,600; 80 / 7,937
May	22 / 2,201; 87 / 9,134
Jun	26 / 2,653; 100 / 9,976
Jul	25 / 2,477; 114 / 10,138
Aug	29 / 2,949; 116 / 10,871
Sep	17 / 1,720; 113 / 11,447

- Checks 5, 6, 10 and 13. Monthly source slices partition identified MAU; do not sum MAU across months. Unidentified sessions excluded from MAU range from 2,286 to 3,703 per month, and are not unique people.

Who drove it

- June to July signups: social 622 to 1,582 (+960), organic 1,305 to 453 (-852), search 700 to 417 (-283), unknown 26 to 25 (-1). These sum to -176, matching total 2,653 to 2,477 (check 14).
- Extreme allocation of all unknowns to a single channel in either month bounds the social change at +934 to +985 and organic change at -878 to -827. Bounds are illustrative and do not allocate real users (check 14).

04. Chapter

Why it happened

04

Hypothesis / status	Evidence and interpretation
H1: Unknown share is large / Rejected for signups	Around 1% each month; 238 of 24,000 total profiles (0.99%) have unknown source. This is not a guarantee of low error in every subgroup (checks 7, 10, 11).
H2: Missing labels can reverse the large July signup lead / Rejected for this example	Even assigning all 25 unknown signups to organic leaves a 1,104 lead for paid social. Close comparisons may behave differently (check 14).
H3: Source label establishes channel causality / Inconclusive	Source is first-touch and no click-level linkage is available; recorded labels may be wrong and confounding is untested. No causal lift or return on spend follows from these counts.

05. Chapter

Data quality and limits

05

Data quality

- Bronze has 24,000 distinct users, 238 literal unknown values, no SQL NULLs or blanks; Silver matches Bronze source values for all profiles (checks 11-12). No flagged Silver profiles; 13 payment corrections appear in quality exceptions (checks 8-9).
- Monthly new-payer unknown counts are 10, 1, 1, 1, 1 and 0; payer denominators can be small, making even one missing label consequential for close CAC comparisons (check 10).
- Gold monthly spend has zero excluded days, but attribution is a channel-label match and cannot verify campaign or click attribution (check 5).

Limitations

- No independent validation of whether recorded known labels are correct; 1% unknown is not the total attribution error rate. Missingness may be systematic within segments.
- The build-time registry SHA-256 (57b21d...1227) differs from this repository copy (4b5b2f...5554), although builder hashes match. Build checks do not demonstrate current definition equivalence or current source freshness.
- Unknown users cannot be assigned without evidence. Sample comparisons are not a universal guarantee for every channel, cohort, period or metric.

06. Chapter

What to do next

06

01 Keep unknown visible, monitor source completeness by signup cohort, platform and campaign, and investigate collection failures.

02 For each decision, bound the result by allocating unknowns adversarially; require stronger evidence when rankings or cost ratios could flip.

03 Audit known source labels against independent attribution records and validate the build registry mismatch and fresh source fingerprints. Use experiments or a defensible causal design for spend decisions.

Appendix

Evidence appendix

A

Checks	Evidence log subjects
01-04	Inventory, snapshot, schema, build metadata and reconciliations
05-06	Marketing and activity monthly source slices
07-09	Bronze values, Silver exceptions and profile coverage
10-12	Gold signup reconciliation, raw completeness, Bronze-Silver agreement
13-14	MAU reconciliation and sensitivity inputs

– Each qNN.jsonl contains SQL, results, limits and timing; qNN-request.json records inputs. All 14 queries succeeded without truncation. Full paths, timestamps, hashes and run count are in RUN-NOTES.md.